

ParkFlow

Smart Parking Management System

System Design Documentation

Data Flow Diagrams (Level 0 / 1 / 2) & Entity-Relationship Diagram

Backend: Node.js · Express · MongoDB (Mongoose)

Frontend: Flutter (offline-first Android client)

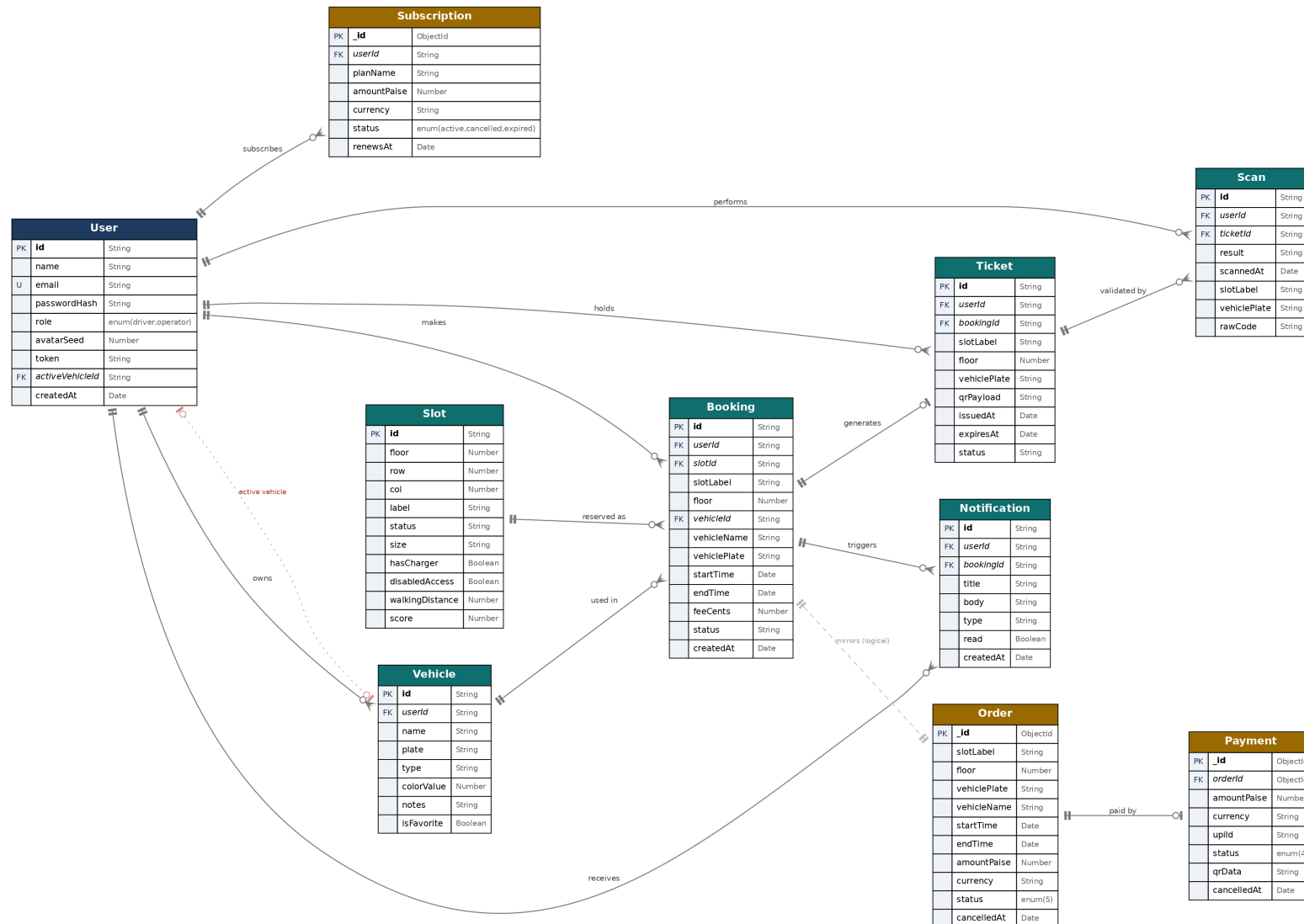
Payments: UPI demo flow · INR (paise) pricing

Generated: 29 June 2026

Contents: ERD · DFD Context · DFD Level 1 · DFD Level 2 (Booking & Payment) · Notation Guide

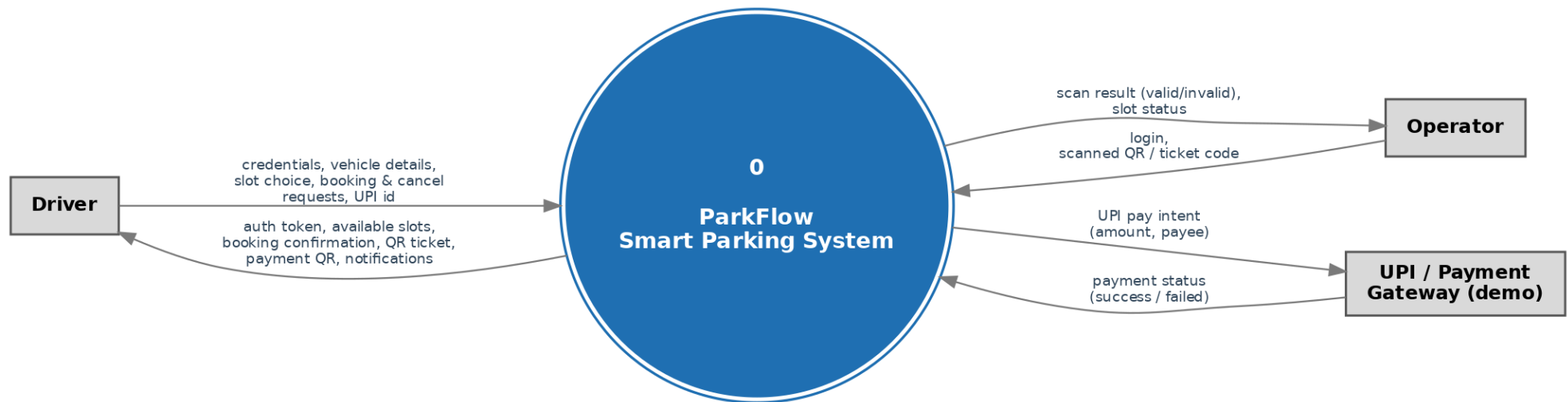
1. Entity-Relationship Diagram (Crow's-Foot)

Logical data model of the ParkFlow database. Ten entities are grouped by colour: User (identity, blue), the offline-first app documents Vehicle/Slot/Booking/Ticket/Scan/Notification (teal), and the commerce records Order/Payment/Subscription (amber). Primary keys are shown in the PK column, foreign keys in the FK column with their data types. Crow's-foot connectors show cardinality (e.g. one User owns zero-or-many Vehicles; each Booking generates at most one Ticket; each Order is paid by at most one live Payment).



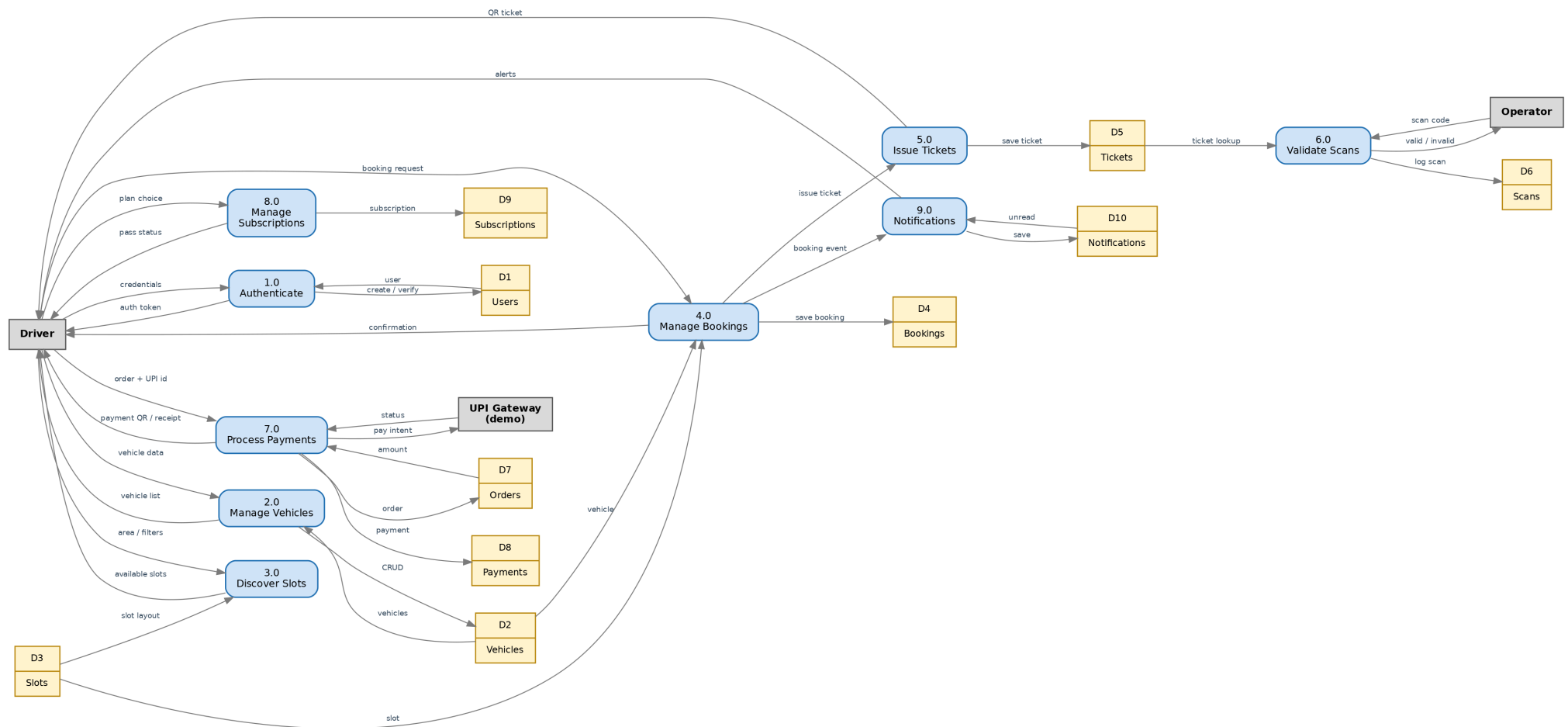
2. DFD — Level 0 (Context Diagram)

The highest-level view: the entire ParkFlow system as a single process (0) with its three external entities — the Driver, the parking Operator, and the UPI / Payment Gateway. Arrows show the net data flows in and out of the system boundary.



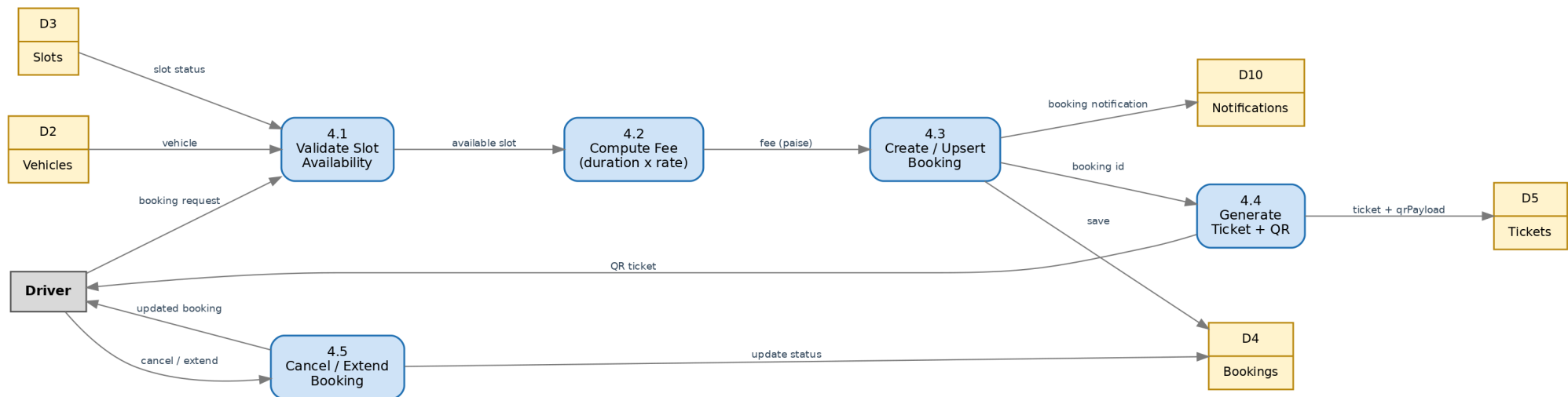
3. DFD — Level 1 (System Decomposition)

Process 0 is decomposed into nine sub-processes (1.0–9.0) and the ten data stores they read from and write to (D1–D10). This maps directly onto the REST API: Authenticate (/api/auth), Vehicles, Slots, Bookings, Tickets, Scans, Payments/Orders, Subscriptions and Notifications. Rounded blue shapes are processes, grey boxes are external entities, and yellow D-tagged shapes are data stores.



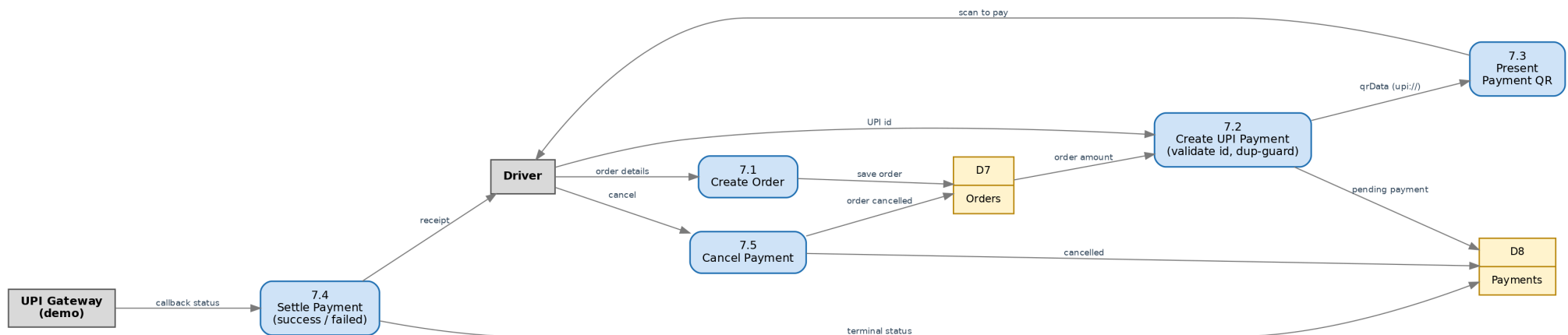
4. DFD — Level 2: Booking Sub-process (explodes 4.0)

Detailed decomposition of Manage Bookings (4.0): validate slot availability against the Slots store, compute the fee from booking duration, create/upsert the Booking, generate the QR Ticket, raise a booking Notification, and handle later cancel/extend requests via PATCH.



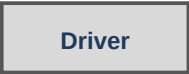
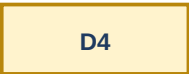
5. DFD — Level 2: Payment Sub-process (explodes 7.0)

Detailed decomposition of Process Payments (7.0): create the Order, create a pending UPI Payment (UPI-id validation plus a database-level duplicate guard so only one live payment exists per order), present the upi:// QR, then settle (success/failed) or cancel the payment and order.



6. Notation Guide

DFD symbols (Gane–Sarson)

	Process — a transformation of data, numbered by level.
	External entity — a data source/sink outside the system boundary.
	Data store — a persisted collection (Dn name).
	Data flow — labelled arrow showing the movement of data.

ERD entity groups & cardinality

	User — identity / account
	App-side offline-first documents (Vehicle, Slot, Booking, Ticket, Scan, Notification)
	Commerce records (Order, Payment, Subscription)

Crow's-foot cardinality

	exactly one
	zero or one
	one or many
	zero or many